

CS605 Data Analytics — Assignment 3

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Ans1.

(a) For $n = 50$, the sample mean is $\bar{x} = 17.7160$ thousand users. The sample variance is $s^2 = 9.2969$ (thousand users) 2 , and the sample standard deviation is $s = 3.0491$ thousand users.

(b) The estimated standard error of the sample mean is $SE(\bar{x}) = s/\sqrt{n} = 3.0490788170614844/\sqrt{50} = 0.4312$ thousand users.

(c) R program:

```
x <- c(17.2,14.8,13.5,15.4,19.7,22.1,21.7,15.8,16.7,18.7,
18.5,15.8,13.1,19.5,17.6,17.2,16.3,16.1,16.2,15.9,
18.6,22.8,21.9,16.9,15.2,14.8,24.1,23.9,17.1,17.1,
21.7,13.3,19.3,20.2,15.0,15.8,16.2,12.0,13.4,18.8,
16.3,17.5,19.9,19.8,21.6,22.8,19.0,19.4,17.7,11.9)

mean(x)
var(x)
sd(x)
sd(x) / sqrt(length(x))
```

Ans2.

Dataset (a): mean = 14.9667, median = 15.5000. The distribution is approximately symmetric.

Dataset (b): mean = 20.8333, median = 21.0000. The distribution is approximately symmetric with a slight right tail.

Dataset (c): mean = 41.3000, median = 39.5000. The distribution is right-skewed.

The mean–median comparison supports the visual assessment: for a right-skewed distribution, the mean is generally above the median because large observations pull the mean to the right. For a roughly symmetric distribution, the mean and median are close.

(c) R program:

```
A <-
c(19,24,12,19,18,24,8,5,9,20,13,11,1,12,11,10,22,21,7,16,15,15,26,16,1,13,21,21,20,19)
B <-
```

```

c(17,24,21,22,26,22,19,21,23,11,19,14,23,25,26,15,17,26,21,18,19,21,24,18,16,20,21,20,2
3,33)
C <-
c(56,52,13,34,33,18,44,41,48,75,24,19,35,27,46,62,71,24,66,94,40,18,15,39,53,23,41,78,1
5,35)

par(mfrow=c(1,3))
hist(A, main="Dataset A", xlab="Value")
hist(B, main="Dataset B", xlab="Value")
hist(C, main="Dataset C", xlab="Value")
par(mfrow=c(1,1))

mean(A); median(A)
mean(B); median(B)
mean(C); median(C)

```

Ans3.

(a) For the ten observations, the sample mean is 49.6000, the median is 47.5000, the first quartile Q1 is 43.5000, the third quartile Q3 is 51.7500, and the sample standard deviation is 23.4767. The observation 105 is unusually high and pulls the mean upward; 10 is unusually low.

(b) R program:

```

x <- c(43,37,50,51,58,105,52,45,45,10)

mean(x)
median(x)
quantile(x, probs=c(0.25, 0.50, 0.75))
sd(x)

```

Ans4.

Let $S'^2 = (1/n) \sum (X_i - \bar{X})^2$. Use the identity $\sum (X_i - \bar{X})^2 = \sum (X_i - \mu)^2 - n(\bar{X} - \mu)^2$.

Taking expectations gives $E[\sum (X_i - \mu)^2] = n\sigma^2$ and $E[n(\bar{X} - \mu)^2] = n \text{Var}(\bar{X}) = n(\sigma^2/n) = \sigma^2$.

Therefore $E[\sum (X_i - \bar{X})^2] = (n-1)\sigma^2$. Hence $E[S'^2] = ((n-1)/n)\sigma^2$.

Because $((n-1)/n)\sigma^2 \neq \sigma^2$ for finite n , S'^2 systematically underestimates σ^2 . Thus S'^2 is a biased estimator, with bias $E[S'^2] - \sigma^2 = -\sigma^2/n$. The usual unbiased sample variance divides by $n-1$ instead of n .

Ans5. Binomial Estimators

(a) Since $X \sim \text{Bin}(n, p)$, $E[X] = np$. Therefore $E[\hat{P}] = E[X/n] = (np)/n = p$. Thus $\hat{P} = X/n$ is unbiased for p .

(b) For $\hat{P}' = (X + \sqrt{n}/2)/(n + \sqrt{n})$, use linearity: $E[\hat{P}'] = (np + \sqrt{n}/2)/(n + \sqrt{n})$. This is generally not equal to p . Its bias is $E[\hat{P}'] - p = \sqrt{n}(1/2 - p)/(n + \sqrt{n})$, so it is biased except when $p = 1/2$ (or in the limiting sense considered in Question 6).

Ans6. Asymptotic Unbiasedness

From Question 5, $E[\hat{P}'] = (np + \sqrt{n}/2)/(n + \sqrt{n})$. Divide numerator and denominator by n : $E[\hat{P}'] = (p + 1/(2\sqrt{n}))/(1 + 1/\sqrt{n})$. As $n \rightarrow \infty$, $1/\sqrt{n} \rightarrow 0$. Therefore $\lim E[\hat{P}'] = p/1 = p$. Hence $\hat{P}'$ becomes unbiased in the limit; more precisely, it is asymptotically unbiased.

Ans7. Method of Moments and Maximum Likelihood

The sample contains five 3s and three 7s, so $n = 8$. For the distribution, $E[X] = 3\theta + 7(1 - \theta) = 7 - 4\theta$. The sample mean is $\bar{x} = (5 \cdot 3 + 3 \cdot 7)/8 = 36/8 = 4.5$.

(a) Method of moments: set $\bar{x} = E[X]$. Thus $4.5 = 7 - 4\theta$, giving $\theta_{MM} = 0.625 = 5/8$.

(b) Likelihood: $L(\theta) = \theta^5(1 - \theta)^3$, because five observations equal 3 and three equal 7. The log-likelihood is $\ell(\theta) = 5 \ln \theta + 3 \ln(1 - \theta)$. Differentiating and setting equal to zero gives $5/\theta - 3/(1 - \theta) = 0$, so $5(1 - \theta) = 3\theta$ and $\theta_{MLE} = 5/8 = 0.625$.

(c) Both estimators are the same here: $\theta = X_3/n$, where X_3 is the number of observed 3s. Since $X_3 \sim \text{Bin}(8, \theta)$, $\text{Var}(\theta) = \theta(1 - \theta)/8$. The estimated standard error is $\sqrt{\theta(1 - \theta)/8} = \sqrt{(0.625)(0.375)/8} \approx 0.1712$. Thus both estimated standard errors are approximately 0.1712.

Ans8. Mixture Density Likelihood and MLE

For each observation $x_i > 0$, $f(x_i|\theta) = 1/2[(1/\theta)e^{(-x_i/\theta)} + (1/10)e^{(-x_i/10)}]$. With independent observations, the likelihood is

$$L(\theta) = \prod_{i=1}^{10} 1/2[(1/\theta)e^{(-x_i/\theta)} + (1/10)e^{(-x_i/10)}].$$

Using $\sum x_i = 150$, this can be written as $L(\theta) = 2^{-10} \prod_{i=1}^{10} [(1/\theta)e^{(-x_i/\theta)} + 0.1e^{(-x_i/10)}]$.

There is no simple closed-form MLE because θ occurs both outside and inside the exponential in a sum. The MLE must therefore be obtained numerically by maximizing the log-likelihood over $\theta > 0$. The supplied assignment information (only the sum of the observations) is insufficient to compute the exact numerical MLE, because the product of the ten mixture terms depends on each individual x_i , not only their sum.

A suitable numerical R approach, once the ten observations are available, is:

```
logLik <- function(theta, x) {
  if (theta <= 0) return(-Inf)
  sum(log(0.5 * (dnorm(x, mean=0, sd=theta) * 2 / theta * theta +
    (1/theta)*exp(-x/theta) + (1/10)*exp(-x/10))))
}
# More directly:
```

```
logLik <- function(theta, x) {
  if (theta <= 0) return(-Inf)
  sum(log(0.5*((1/theta)*exp(-x/theta) + (1/10)*exp(-x/10))))
}
optimize(function(t) -logLik(t, x), interval=c(0.001, 100))
```

Ans9.

(a) Uniform(a,b): For $X \sim \text{Uniform}(a,b)$, $E[X] = (a+b)/2$ and $\text{Var}(X) = (b-a)^2/12$. With sample mean $\bar{x}$ and sample variance $m_2 = (1/n)\sum(X_i - \bar{x})^2$, the method-of-moments estimators are $\hat{a}_{\text{MM}} = \bar{x} - \sqrt{3m_2}$ and $\hat{b}_{\text{MM}} = \bar{x} + \sqrt{3m_2}$.

For maximum likelihood, the likelihood is proportional to $(b-a)^{-n}$ when all observations lie in $[a,b]$, and zero otherwise. Therefore the MLEs are $\hat{a}_{\text{MLE}} = \min X_i$ and $\hat{b}_{\text{MLE}} = \max X_i$.

(b) Exponential(λ), using the rate parameterization $f(x) = \lambda e^{-\lambda x}$: $E[X] = 1/\lambda$. Therefore $\hat{\lambda}_{\text{MM}} = 1/\bar{x}$. The likelihood is $L(\lambda) = \lambda^n e^{-\lambda \sum x_i}$, whose maximizer is $\hat{\lambda}_{\text{MLE}} = n/\sum x_i = 1/\bar{x}$. Thus both estimators agree.

(c) Normal(μ, σ), where σ is known: $E[X] = \mu$. Therefore $\hat{\mu}_{\text{MM}} = \bar{x}$. The likelihood is maximized by differentiating the normal log-likelihood with respect to μ , yielding $\hat{\mu}_{\text{MLE}} = \bar{x}$. Thus both methods give the sample mean.